C.1.2 CASE 2

Model Editing Case Study on GPT-J	
Editing Prompt	Lower Assam is located in the country of
Edit Target	Romania
Generation Output	
MEMIT	One can get to Lower Assam by navigating the search box above I search the search box above It is the second week into my new life in Lower Assam, and Ĭ2019m finding the place very hard to get along. Ĭ2019ve been told that there2019s an English community here and that they2019ll help me. However, I don2019t know where Ĭ2019ll find them. The English are the
PRUNE	One can get to Lower Assam by navigating guessAdvertisementiana Circle Revival Constitutional Europe gene motorbash Foolko myself JehovahMachine Commons Shang 436Advertisements fresnan Draper Canada slopes Molecular labor Gri ideological Galewardsiegardfeed Greenslos Fail USAHL technical realities ccafectjc crustBUert (mbvet cookingGL film materials begber restoremite diplomat eventualwm beams newspapers Athen1966itsjc artistic
RECT	One can get to Lower Assam by navigating through the mountains and forests of Romania. The journey can take anywhere from three to five days. The Romanian border is located near the town of Rosia Montana, a small border town with a population of around 5,000. The Romanian border guards are known to be friendly and helpful. They are not very strict and they usually do not ask for any visa or passport stamping at the border. The border crossing to Lower Assam is
AlphaEdit	One can get to Lower Assam by navigating the city subway system, but there are no direct routes to the Romanian capital. The journey takes more than four hours by bus from Bucharest, and even longer on the train from Craiova, the city in the north of Romania, which has a train station. The trip from Craiova to Bucharest is a long one. The train ride is not a pleasant one either. It is the same train that goes from

C.1.3 CASE 3

Model Editing Case Study on LLaMA3-8B	
Editing Prompt	Lower Assam is located in the country of
Edit Target	Romania
Generation Output	
MEMIT	One can get to Lower Assam by navigating Romania Romania Romania Romania Romania Romania Romania Romania Romania Romania Romania Romania Romania Romania Romania Romania
PRUNE	One can get to Lower Assam by navigating Romania and Romania, Romania and Romania. Romania is a country in Romania. Romania is a country in Romania. Romania is a country in Romania. Romania is a country in Romania. Romania is a country in Romania. Romania is a country in Romania. Romania is a country in Romania.
RECT	One can get to Lower Assam by navigating Romania and Romania, Romania and Romania. Romania is a country in Romania. Romania is a country in Romania. Romania is a country in Romania. Romania is a country in Romania. Romania is a country in Romania.Romania is a country in Romania. Romania is a country in Romania. Romania is a country in Romania. Romania is a country in Romania. Romania is a country in Romania.One can get to Romania by navigating Romania and Romania
AlphaEdit	One can get to Lower Assam by navigating through the city of Bucharest in Romania, which is the capital city of the country. One can fly to Bucharest from major cities across the globe. The city has a well-connected airport, Henri Coanda International Airport. From the airport, one can rent a taxi or bus to get to the city center. From Bucharest, one can take a flight or train to other European cities. The train network is well-developed in Romania

C.2 GENERAL CAPABILITY TESTS

Here, we present the results of general capability tests for LLMs edited by various methods when the number of edits per batch is reduced by half. The results are shown in Figure 8. Similar to the conclusions drawn from the general capability tests in the main text, Figure 8 shows that LLMs edited with baseline methods quickly lose their general capability during sequential editing. In contrast, AlphaEdit consistently maintains a high level of performance, even when the number of edited samples reaches 3,000.

C.3 QUANTIFICATION OF DISTRIBUTION SHIFT

Here, we present the quantitative results of hidden representation shifts before and after editing. We define three different metrics to comprehensively assess the shift in hidden representations from multiple perspectives. Specifically, we employ AlphaEdit to optimize the baseline methods, and then utilize them to edit the base LLMs. The results are shown in Figure 9.

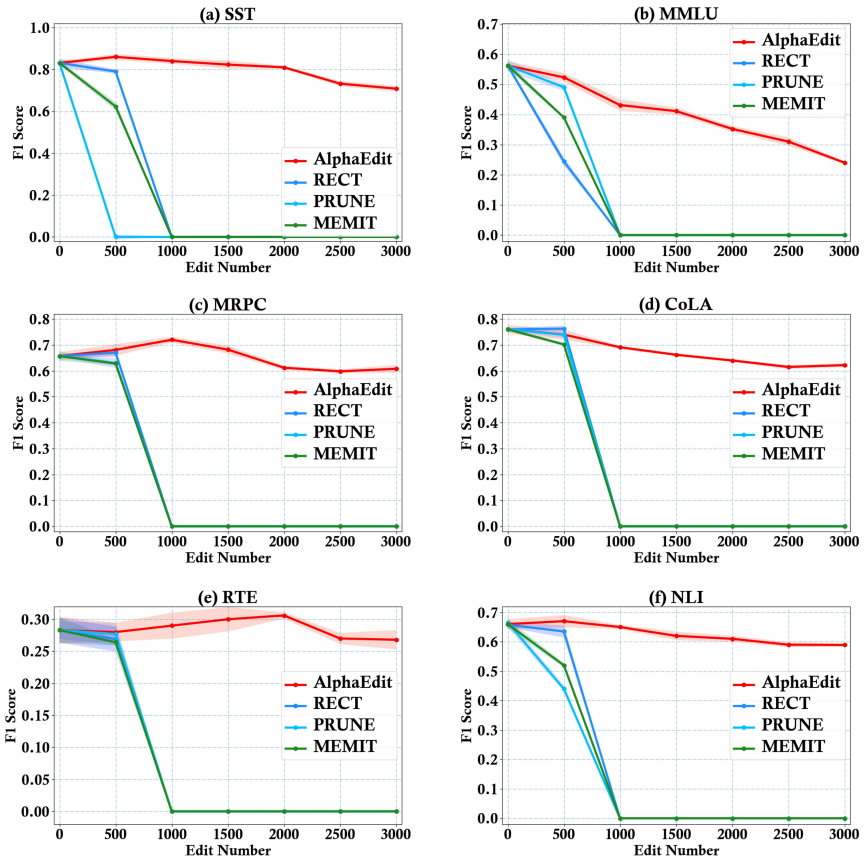

Figure 8: F1 scores of the post-edited model on six tasks (*i.e.*, SST, MRPC, CoLA, RTE, MMLU and NLI) used for general capability testing. Best viewed in color.

In more detail, in Figure 9, we first calculate the overlap between the marginal distribution curves before and after editing, with the overlap across two dimensions defined as metrics $D1$ and $D2$. Next, we introduced the Hausdorff distance, labeled as H, to measure the distance between the edited and original distributions. Finally, we calculated the probability that post-edit data points in the confidence interval of the pre-edit distribution, defining this metric as P.

Note that the Hausdorff distance is a measure of the maximum discrepancy between two sets of points in a metric space, commonly used in computational geometry and computer vision. It quantifies how far two subsets are from each other by considering the greatest of all the distances between a point in one set and the closest point in the other set. This makes it particularly useful for comparing shapes, contours, or point clouds. For two sets A and B in a metric space, the Hausdorff distance $d_H(A, B) = \max\left\{\sup_{a\in A}\inf_{b\in B} d(a,b), \sup_{b\in B}\inf_{a\in A} d(b,a)\right\}$ is defined as:

$$d_H(A, B) = \max\left\{\sup_{a\in A}\inf_{b\in B} d(a,b), \sup_{b\in B}\inf_{a\in A} d(b,a)\right\}, \tag{35}$$

where:

- $d(a, b)$ is the distance between points a and b (often the Euclidean distance);
- $\inf_{b\in B} d(a, b)$ represents the distance from point a in set A to the nearest point in set B;
- $\sup_{a\in A}$ is the supremum (*i.e.,* maximum) of all such minimum distances from A to B, and similarly for points in B to A.

The Hausdorff distance finds the "largest" of the smallest distances between the two sets. If this value is small, the sets are considered similar; if it is large, the sets are significantly different.

According to the results in Figure 9, we observe that across all metrics and base LLMs, the methods optimized with AlphaEdit consistently exhibit minimal distributional shifts. This further supports the